

Boundary Cases of the 4×4 SAP–Lunar Problem

Automated mathematics run `fa_banach_001`
Candidate partial result for expert review

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Status. Candidate partial result, likely valid. The five-color case is computer-assisted with exact arithmetic; the high-color cutoff is a short combinatorial proof. The original problem remains open here for six through thirteen colors.

1 Source problem

Han, Qiu, and Wang ask the following in Problem 7 of [1], on PDF page 21.

Consider any map $\Phi : \{1, 2, 3, 4\} \times \{1, 2, 3, 4\} \rightarrow \mathcal{L}$. Assume that the Hankel system associated to Φ has SAP. Does it follow that Φ is a lunar map?

Definition 4 (Lunar maps). Consider three sets $\mathcal{A}, \mathcal{X}, \mathcal{L}$. A two-variable map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$ is called a *lunar map* if

- (i) Φ is *coordinatewise injective*, that is, the map $\Phi(a, \cdot)$ is injective on $\mathcal{X}$ for each $a \in \mathcal{A}$ and so is the map $\Phi(\cdot, x)$ on $\mathcal{A}$ for each $x \in \mathcal{X}$.
- (ii) Φ satisfies the *generalized lunar condition*: for any $a, b, c, d \in \mathcal{A}$ and $x, y, z, w \in \mathcal{X}$, the following implication holds:

$$(5.1) \quad \left(\Phi(a, x) = \Phi(b, y), \Phi(c, x) = \Phi(d, y), \Phi(a, z) = \Phi(b, w) \right) \implies \left(\Phi(c, z) = \Phi(d, w) \right).$$

Given a coordinatewise injective map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$, by the Hankel system associated to the map Φ , we mean the family $\{\Gamma_\ell^\Phi\}_{\ell \in \mathcal{L}} \subset B(\ell^2(\mathcal{X}), \ell^2(\mathcal{A}))$ consisting of operators Γ_ℓ^Φ :

$$(5.2) \quad \Gamma_\ell^\Phi(a, x) = \mathbf{1}(\Phi(a, x) = \ell), \quad (a, x) \in \mathcal{A} \times \mathcal{X}.$$

Note that Φ is coordinatewise injective if and only if the operators $\Gamma_\ell^\Phi \in B(\ell^2(\mathcal{X}), \ell^2(\mathcal{A}))$ are of unit norm $\|\Gamma_\ell^\Phi\| = 1$ for all ℓ in the image of Φ .

Theorem B. For any lunar map $\Phi : \mathcal{A} \times \mathcal{X} \rightarrow \mathcal{L}$, the Hankel system $\{\Gamma_\ell^\Phi\}_{\ell \in \mathcal{L}}$ has SAP.

One may wonder whether the lunar condition on a two-variable map Φ is also necessary for the Hankel system $\{\Gamma_\ell^\Phi\}_\ell$ to have SAP. However, even for maps on finite sets of very small size, we are not able to answer this question.

Problem 7. Consider any the map $\Phi : \{1, 2, 3, 4\} \times \{1, 2, 3, 4\} \rightarrow \mathcal{L}$. Assume that the Hankel system of associated to Φ has SAP. Does it follow that Φ is a lunar map ?

5.1. Hereditary SAP. In general, the SAP is not preserved by compression (see Example 3.3).

Figure 1: The definition, preceding theorem, and Problem 7 in the source paper, PDF page 21.

Let $[4] = \{1, 2, 3, 4\}$. Coordinatewise injectivity means that no color $\ell \in \mathcal{L}$ occurs twice in a row or twice in a column. Write

$$\Gamma_\ell(a, x) = \mathbf{1}_{\{\Phi(a, x) = \ell\}}.$$

For scalar coefficients c_ℓ , set

$$A(c) = [c_{\Phi(a,x)}]_{a,x \in [4]}, \quad B(c) = \sum_{\ell} c_{\ell} \Gamma_{\ell} \otimes \Gamma_{\ell}.$$

The upper SAP inequality, even just for scalar coefficients, requires

$$\|B(c)\| \leq \|A(c)\| \quad \text{for every } c. \quad (1)$$

The diagonal row-pair and column-pair block of $B(c)$ is a copy of $A(c)$, so the reverse inequality is automatic.

The generalized lunar condition is

$$\begin{aligned} \Phi(a, x) = \Phi(b, y), \quad \Phi(c, x) = \Phi(d, y), \quad \Phi(a, z) = \Phi(b, w) \\ \implies \Phi(c, z) = \Phi(d, w). \end{aligned}$$

2 Claimed partial result

Theorem 1 (Boundary color counts). *Let $\Phi : [4] \times [4] \rightarrow \mathcal{L}$ be coordinatewise injective and let $m = |\text{im } \Phi|$.*

1. *If $m \leq 5$, then the associated Hankel system has SAP if and only if Φ is lunar.*

2. *If $m \geq 14$, then Φ is lunar, without any SAP assumption.*

Consequently, Problem 7 has a positive answer for $m \leq 5$ and for $m \geq 14$. Only $6 \leq m \leq 13$ remains open in this packet.

The forward implication “lunar implies SAP” is Theorem B of [1]. The new content is the converse at $m \leq 5$ and the automatic lunar property at $m \geq 14$.

3 Proof intuition

A coordinatewise injective table is a proper edge-coloring of the bipartite graph $K_{4,4}$: every color class is a matching. At the low-color boundary, there are few enough such partitions to classify exactly. After quotienting by the evident symmetries, only sixteen non-lunar five-color patterns remain. Each admits a tiny integer scalar substitution that makes an off-diagonal tensor block larger than the original matrix. Exact Sturm counts, rather than decimal singular values, certify every strict inequality.

At the opposite boundary, failure of the lunar implication already forces three independent pairs of cells to have equal colors. Starting from sixteen independent cells, these identifications leave at most thirteen color classes. Thus fourteen colors rule out non-lunarity before operator norms enter.

4 The high-color argument

Lemma 1. *If a coordinatewise injective 4×4 map is non-lunar, then it uses at most thirteen colors.*

Proof. Choose a failed lunar implication and form three equality edges on the set of sixteen cells:

$$e_1 = \{(a, x), (b, y)\}, \quad e_2 = \{(c, x), (d, y)\}, \quad e_3 = \{(a, z), (b, w)\}.$$

Each edge joins equal-colored cells. In a failed implication these are non-loop edges. Indeed, a loop in e_1 or e_2 , together with coordinatewise injectivity, collapses both row pairs; a loop in e_3 first collapses the first row pair. In every case the proposed conclusion then follows.

Moreover, e_1 is vertex-disjoint from e_2 . A common endpoint in the same column forces $a = c$ or $b = d$ and then forces the other row equality by column injectivity; a crossed common endpoint forces $x = y$, which similarly makes the conclusion automatic. The same argument, using row injectivity, shows that e_1 is vertex-disjoint from e_3 .

Finally, e_2 and e_3 are distinct. If they agree with the same orientation, the desired conclusion is the first premise; if they agree with opposite orientation, it is again the first premise with the two sides reversed. Hence the graph with edges e_1, e_2, e_3 has equivalence-rank at least three: e_1 contributes one merger and the two distinct non-loop edges e_2, e_3 contribute two more. Therefore the color partition of the sixteen cells has at most $16 - 3 = 13$ classes. $\square$

The second assertion of Theorem 1 is the contrapositive of Lemma 1.

5 Exact finite proof at five colors

5.1 Complete enumeration

The verifier represents a table by a restricted-growth word of length sixteen. At each cell it either reuses a previous color that has appeared in neither the current row nor the current column, or introduces the next new color. Every partition of the cells into matchings has exactly one such labeling, so the recursion is exhaustive and duplication-free.

The exact census is:

colors	canonical tables	lunar	non-lunar
4	24	24	0
5	4896	144	4752

The code then takes the minimum restricted-growth word over all row and column permutations, color relabelings, and transposition. Transposition is harmless here because a scalar norm witness transposes both $A(c)$ and $B(c)$. The five-color tables yield seventeen isomorphism classes: one lunar and sixteen non-lunar.

5.2 Exact norm certificates

For each non-lunar representative, the verifier searches primitive vectors $c \in \{-3, -2, \dots, 3\}^5$. A decimal singular-value computation only proposes a candidate. Acceptance is exact.

Let

$$p_A(t) = \det(tI - A(c)^*A(c)), \quad p_B(t) = \det(tI - B(c)^*B(c)).$$

Both are integer polynomials with only real nonnegative roots. The verifier chooses a rational q between the proposed squared norms and uses exact Sturm root counts to prove

$$\#\{\lambda \in \text{roots}(p_A) : \lambda > q\} = 0, \quad \#\{\lambda \in \text{roots}(p_B) : \lambda > q\} > 0.$$

Thus $\|A(c)\|^2 < q < \|B(c)\|^2$ exactly, contradicting (1). The sixteen coefficient certificates are:

class	coefficient vector	numerical ratio (display only)
0	(0, 1, -3, -3, -3)	1.092280
1	(1, -3, -3, -3, 0)	1.047174
2	(0, 1, -3, -3, -3)	1.044094
3	(1, -3, -3, 2, 3)	1.003121
4	(0, 1, -3, -3, -3)	1.003987
5	(0, 1, -3, -3, -3)	1.047174
6	(0, 1, -3, -3, -3)	1.010169
7	(0, 1, -3, 1, -3)	1.016851
8	(0, 0, 1, -3, -1)	1.011773
9	(1, -3, -3, -3, 1)	1.025126
10	(0, 0, 1, -3, -3)	1.043578
11	(0, 0, 1, -3, -3)	1.043578
12	(0, 1, -3, 0, 2)	1.031391
13	(0, 1, -3, 1, 1)	1.037650
14	(0, 1, 1, 1, -1)	1.154701
15	(0, 0, 1, -1, 2)	1.043068

The representative tables, rational thresholds, and exact root counts are regenerated by the supplied verifier rather than trusted from this display. This proves the new converse for five colors. Together with the four-color census and Theorem B of [1], it proves the first assertion of Theorem 1.

6 Reproducibility and verification

Run from the packet directory:

```
conda run --no-capture-output -n sandbox python \
  code/verify_five_color_boundary.py
```

The script checks all census and orbit counts, finds all sixteen certificates, and ends with

PASS: every non-lunar five-color class fails scalar SAP exactly

An adversarial review found and repaired one bug during development: the first canonical relabeler wrote through a flattened view that became a copy after transposition. No class count from that run was retained. The final version constructs a fresh contiguous output array, and every count and certificate in this packet was recomputed afterward. Independent exploratory matrix-valued searches were not used in the proof.

7 Limitations and novelty check

This is a partial result. It does not decide the source problem for $6 \leq |\text{im } \Phi| \leq 13$. It also does not replace the source authors’ general theorem that lunar maps have SAP.

A bounded novelty search on 21 July 2026 used the exact paper title, exact Problem 7 wording, the phrases “lunar map”, “Hankel SAP converse”, and the author names in arXiv and general web search. It found arXiv:2407.01048v1 and a contemporaneous 2024 talk announcement describing unspecified progress, but no later paper or public theorem resolving these boundary cases. Novelty confidence is therefore medium, not high, until the authors or an expert check for unpublished overlap.

8 Human-review recommendation

The result is suitable for expert review. The two priority checks are:

1. audit that the restricted-growth recursion and orbit minimization cover exactly the claimed tables; and
2. audit one or two exact Sturm certificates, especially class 3, which has the smallest displayed margin.

References

- [1] Y. Han, Y. Qiu, and Z. Wang, *Self-absorption of Hankel systems on monoids – a seemingly universal property*, arXiv:2407.01048v1, 2024.